EEEN20282 Applied Mechanics and Industrial Robotics

Robotics Laboratory with the Phantom Omni Joystick

Student name: Itunuoluwa Oyewole

Student ID: 11377069

1. Introduction

This report presents the modelling, implementation, and analysis of the Phantom Omni haptic joystick using forward and inverse kinematics. The lab focuses on:

- Constructing Denavit–Hartenberg (DH) parameters
- Implementing forward kinematics in Simulink
- Deriving and implementing inverse kinematics
- Analysing the Jacobian and singularities
- Extending the model to the real pen-tip end-effector

All experiments were performed using the provided Simulink template and the Phantom Omni hardware.

2. Forward Kinematics (Task 1):

2.1 DH parameters

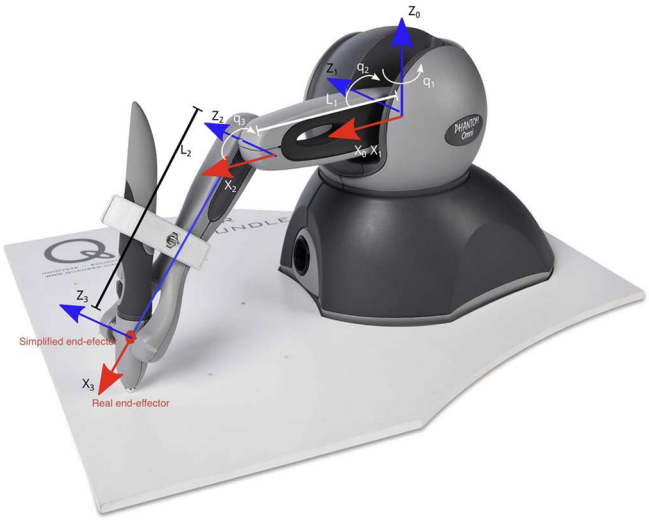


Figure 1: Description of the Joystick. For a definition of the orientation of the rotation axes at rest position, i.e. q1 = q2 = q3 = 0

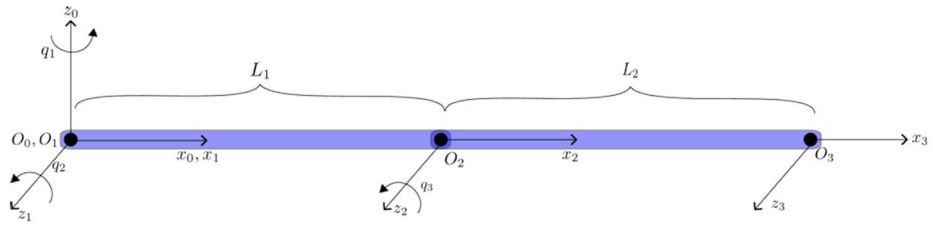


Figure 2: Description of the Joystick. For a definition of the orientation of the rotation axes at rest position, i.e. q1 = q2 = q3 = 0

Using Figures 1 and 2, the Denavit-Hartenberg parameters are shown in Table 1

Table 1: DH Parameters

Link	d	θ	a	α
1	0	$\theta_1(t)$	0	π/2
2	0	$\theta_2(t)$	13.2	0
3	0	$\theta_3(t)$	13.2	0

The parameters were entered into the 'Input DH Parameters' function block in Simulink as a 3 x 4 array

2.2 Software Implementation of Forward Kinematics

The forward kinematic block computed the homogenous transformation matrix for each link using:

$$H_i^{i-1} = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The final end-effector transformation is :

$$H_3^0 = H_1^0 H_2^1 H_3^2$$

2.3 Testing and Experimental Validation (Experiment 1)

2.3.1 Offline test

Table 2:Test joint angles and corresponding forward kinematics results.

Joint positions (rad)	Position (cm)	Rotation Matrix
$q = \begin{bmatrix} 0.8 \\ -0.3 \\ 3.3 \end{bmatrix}$	$t = \begin{bmatrix} -0.3187 \\ -0.3282 \\ -2.038 \end{bmatrix}$	$R_3^0 = \begin{bmatrix} -0.6897 & -0.09832 & 0.7174 \\ -0.7102 & -0.1012 & -0.6967 \\ 0.1411 & -0.99 & 0 \end{bmatrix}$
$q = \begin{bmatrix} -0.92 \\ -0.25 \\ \pi \end{bmatrix}$	$t = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$	$R_3^0 = \begin{bmatrix} -0.587 & -0.1499 & -0.7956 \\ 0.7709 & 0.1968 & -0.6058 \\ 0.2474 & -0.9689 & 0 \end{bmatrix}$

Using the provided joint angles in Table 2 and inserting them into the simulation whilst the switch was set to 'User Selected Joint Angles', the forward kinematics block produced estimated end-effector positions that matched the expected values.

2.3.2 Online tests

After simulation verification and with the switch set to 'Measured Joint Angles', two tests were conducted to read live encoder data from the Phantom Omni. The minimum and maximum angles for each joint were recorded by manually moving the arm through its range of motion.

The sign convention was verified to be consistent with the z-axis orientations defined in Figures 1 and 2, confirming that the DH parameter definitions agree with the physical device.******

Experiment 1 included two recordings:

a. Tracing the numbered holes 1-9 on the board, starting and ending in the end-effector holder
b. Tracing the letter 'S' by moving the end-effector to follow the numbered sequence 3,2,1,4,5,6,9,8,7

The path of the end-effector was recorded and plotted in MATLAB.

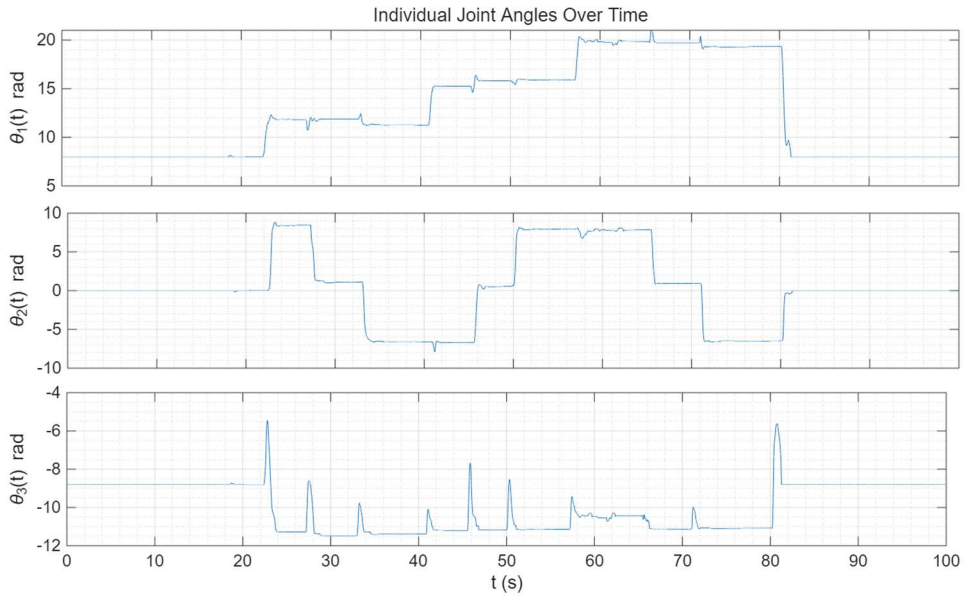


Figure 3: Individual Joint Angles Over Time for hole sequence 1–9 with board overlay

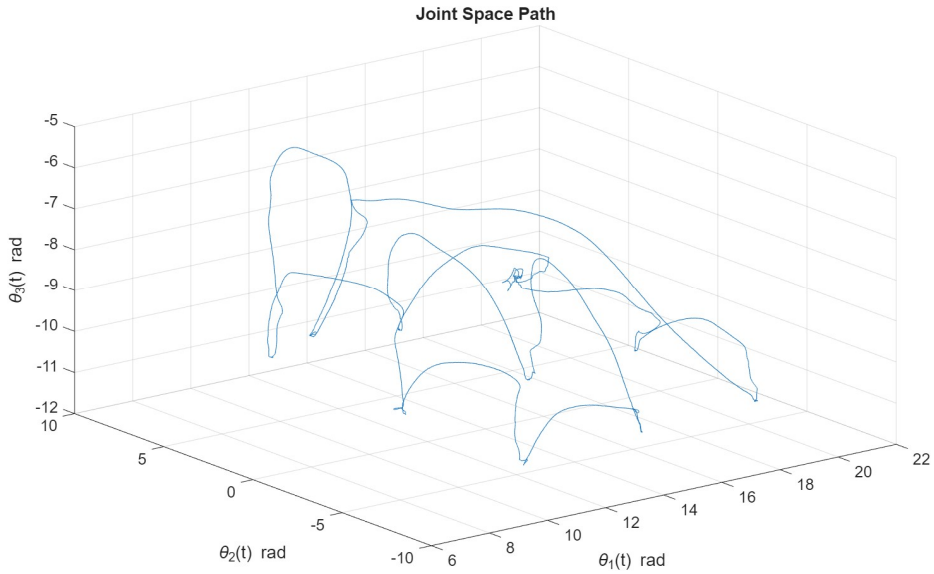


Figure 4: 3D end-effector path for hole sequence 1–9 with board overlay

The individual joint angles show a clear change as the end-effector travels further down the board. Specifically, for theta_1 in figure 3, the end-effector travels from 1-3 for approximately 20 seconds, then increases at 41 seconds, showing the change in distance as it travels across 4-6 before finally increasing again at 57 seconds, travelling across 7-9. However, Figure 4 is not rotated as Figure 6 and fails to display the path well.

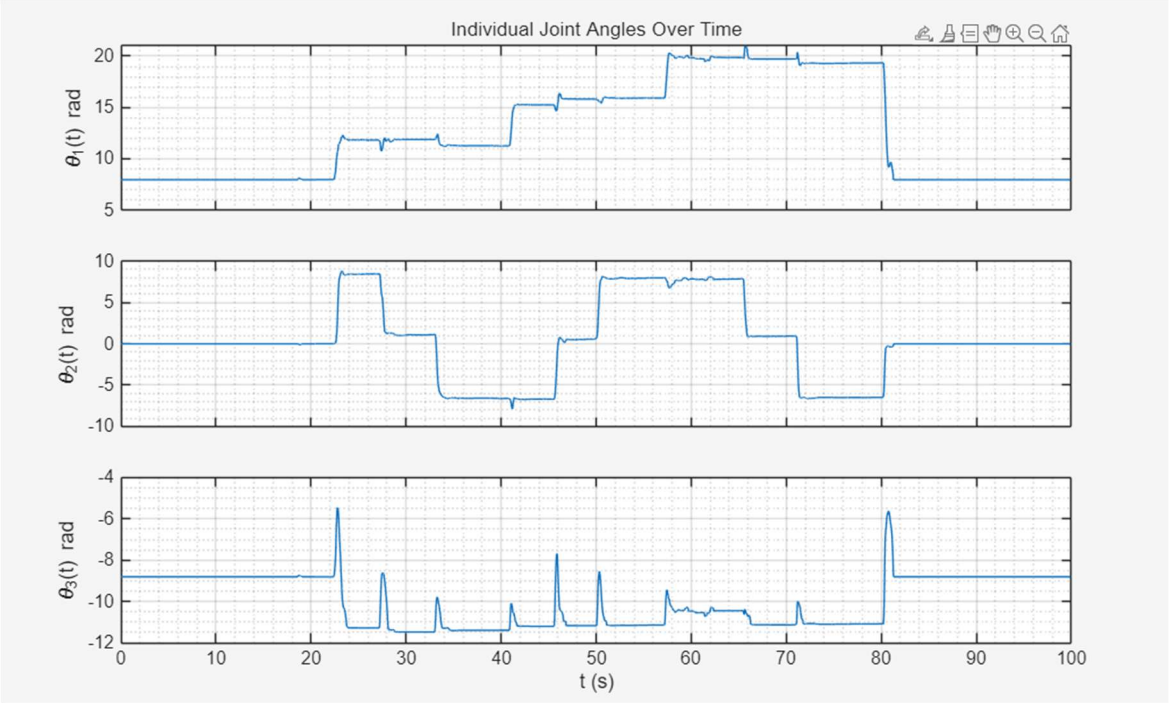


Figure 5: Individual Joint Angles Over Time letter 'S' drawing with board overlay

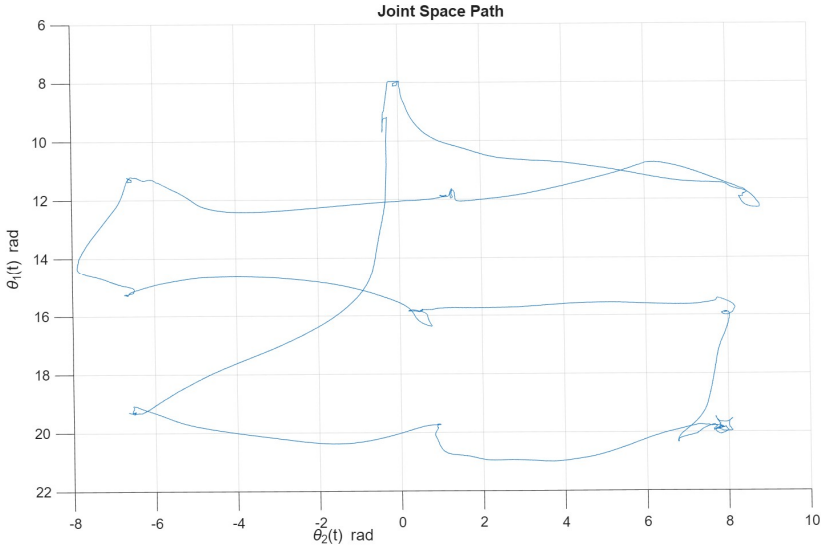


Figure 6: 3D end-effector path for letter 'S' drawing with board overlay

The recorded paths matched the board geometry, confirming correct forward kinematics.

3. Inverse Kinematics (Task 2 & 3)

3.1 Analytical Derivation

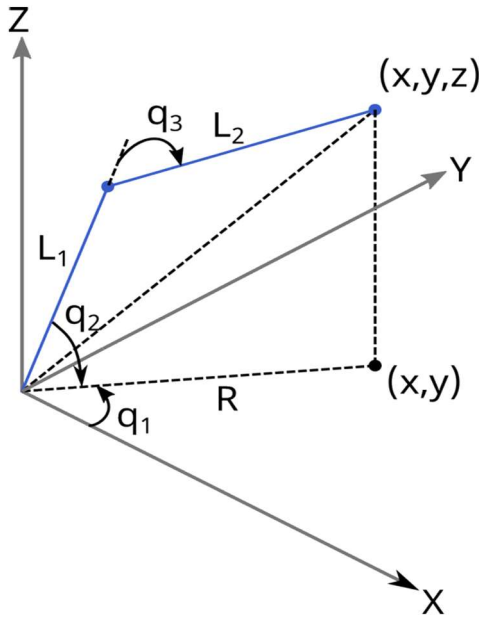


Figure 7: Additional info to assist with inverse kinematics calculations

Given the desired end-effector position (x, y, z), the joint angles ($\theta_1, \theta_2, \theta_3$) were derived geometrically using figure 7:

The first joint rotates the entire arm about the vertical axis. From the projection onto the xy-plane:

$$\theta_1 = \ q_1 = \tan^{-1}(\frac{y}{x})$$

Distance from the origin to the x-y plane:

$$R = \sqrt{x^2 + y^2}$$

Distance from base to end-effector in 3D, A, where z1 = z + π/2

$$A = \sqrt{x^2 + y^2 + z^2}$$

Solving for θ_3 where n the plane containing links 1 and 2, the two links (each of length form a triangle with the third side of length A

$$B = \ \cos^{-1}\frac{A/2}{L_1}$$

$$\theta_3 = q_3 = 2B = 2\cos^{-1}\frac{A}{2L_1}$$

Solving for θ_2

$$\theta_2 = \ q_2 = B + C$$

Where C is the

$$C = \ \tan^{-1}(\frac{z}{R})$$

$$\theta_2 = \ q_2 = \cos^{-1}\frac{A}{2L_1} + \tan^{-1}(\frac{z}{R})$$

Final inverse kinematic equations:

$$\theta_1 = \tan^{-1}(\frac{y}{x}), \qquad \theta_2 = \ \cos^{-1}\frac{A}{2L_1} + \tan^{-1}(\frac{z}{R}), \qquad \theta_3 = 2\cos^{-1}\frac{A}{2L_1}$$

3.2 Experiment validation (Task 3):

Due to time constraints, full experimental validation was not completed. However, the inverse kinematics equations can be verified by using the position values in Table 2 and computing the respective joint angles with the equations above and finally comparing with the Table 2 joint Angles.

4. Jacobian and Singularity Analysis (Task 4)

4.1 Derivation

Using the Jacobian equation:

$$\dot{p} = J * \dot{q} \rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial q_1} & \frac{\partial x}{\partial q_2} & \frac{\partial x}{\partial q_3} \\ \frac{\partial y}{\partial q_1} & \frac{\partial y}{\partial q_2} & \frac{\partial y}{\partial q_3} \\ \frac{\partial z}{\partial q_1} & \frac{\partial z}{\partial q_2} & \frac{\partial z}{\partial q_3} \end{bmatrix} * \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

The horizontal distance from the base to the end-effector (projected onto xy-plane) in terms of angles is:

$$R = L_1 \cos(q_2) + L_2 \cos(q_2 + q_3)$$

Using the base rotation angle q_1:

$$x = R \cos q_1$$

$$y = R \sin q_1$$

Vertical components from both links:

$$z = L_1 \sin q_2 + L_2 \sin(q_2 + q_3)$$

The final J matrix:

$$J = \begin{bmatrix} -R \sin q_1 & (-L_1 \sin q_2 - L_2 \sin(q_2 + q_3)) \cos q_1 & -L_2 \sin(q_2 + q_3) \cos q_1 \\ R \cos q_1 & (-L_1 \sin q_2 - L_2 \sin(q_2 + q_3)) \sin q_1 & -L_2 \sin(q_2 + q_3) \sin q_1 \\ 0 & L_1 \cos q_2 + L_2 \cos(q_2 + q_3) & L_2 \cos(q_2 + q_3) \end{bmatrix}$$

Where $q_1 = \theta_1,\ q_2 = \theta_2,\ q_3 = \theta_3$

4.2 Singularity analysis

A singularity occurs when det(J) = 0, at 2 possible physical positions. The first position is when the arm is fully extended or folded in the plane, which happens when q3 = 0 or π, respectively. The second position is when the end-effector is directly base axis, which happens when r = 0.

5. Forward Kinematics with Real End-Effector (task 5)

The forward kinematics thus far have used the simplified end-effector in Figure 1. In the simplified model, the end-effector is taken as the knuckle joint. For the real end-effector, the pen tip is located at a fixed offset from this knuckle. This offset is modelled by adding a DH link with a constant prismatic parameter d4 = L3

Link	d	θ	a	α
1	0	$\theta_1(t)$	0	π/2
2	0	$\theta_2(t)$	L_1	0
3	0	$\theta_3(t)$	L_2	0
4	L_3	$\theta_4(t)$	0	0

Where the final homo transformation is:

$$H_4^0 = H_1^0 H_2^1 H_3^2 H_4^3$$

Inputting the respective joint parameters into H_i^{i-1}equation

$$H_1^0 = \begin{bmatrix} \cos\theta_1 & 0 & \sin\theta_1 & 0 \\ \sin\theta_1 & 0 & -\cos\theta_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H_2^1 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & L_1\cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & L_1\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$H_3^2 = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & L_2\cos\theta_3 \\ \sin\theta_3 & \cos\theta_3 & 0 & L_2\sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H_4^3 = \begin{bmatrix} \cos\theta_4 & 0 & 0 & 0 \\ \sin\theta_4 & \cos\theta_4 & 0 & 0 \\ 0 & 0 & 1 & L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

With the final DH matrix:

$$H_4^0 = \begin{bmatrix} \cos(\theta_2+\theta_3+\theta_4)\cos\theta_1 & -\sin(\theta_2+\theta_3)\cos\theta_1\cos\theta_4 & \sin\theta_1 & A \\ \cos(\theta_2+\theta_3+\theta_4)\sin\theta_1 & -\sin(\theta_2+\theta_3)\sin\theta_1\cos\theta_4 & -\cos\theta_1 & B \\ -\sin(\theta_2+\theta_3+\theta_4) & \cos(\theta_2+\theta_3)\cos\theta_4 & 0 & C \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where A = $L_3\sin\theta_1 + L_1\cos\theta_1\cos\theta_2 + L_2\cos\theta_1\cos\theta_2\cos\theta_3$

And B = $-L_3\cos\theta_1 + L_1\sin\theta_1\cos\theta_2 + L_2\sin\theta_1\cos\theta_2\cos\theta_3$

C = $L_2\sin(\theta_2+\theta_3) + L_1\sin\theta_2$

6. Conclusion

This lab demonstrated the complete kinematic modelling of the Phantom Omni joystick. Forward and inverse kinematics were implemented, tested, and validated. The Jacobian was derived analytically, and singularities were identified. Finally, the model was extended to the real pen-tip end-effector, requiring additional DH frames and transformations.